DSA ki Baithak

CODE BAITHAK

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Teaser / Trailer :

```
*   *   *
  *   *
*   *
  *
*
```

```
* * * * *
  * * * *
    * * *
      * *
        *
      * *
    * * *
  * * * *
* * * * *
```

```
    *
  *   *
*       *
*       *
  *   *
    *
```

```
4           4
4 3       3 4
4 3 2     2 3 4
4 3 2 1   1 2 3 4
4 3 2 1 0 1 2 3 4
4 3 2 1   1 2 3 4
4 3 2     2 3 4
4 3       3 4
4           4
```

```
        1
      2 1   1 2
    3 2 1     1 2 3
4 3 2 1         1 2 3 4
    3 2 1     1 2 3
      2 1   1 2
        1
```

```
5 5 5 5 5 5 5 5 5
5 4 4 4 4 4 4 4 5
5 4 3 3 3 3 3 4 5
5 4 3 2 2 2 3 4 5
5 4 3 2 1 2 3 4 5
5 4 3 2 2 2 3 4 5
5 4 3 3 3 3 3 4 5
5 4 4 4 4 4 4 4 5
5 5 5 5 5 5 5 5 5
```

"Patterns Ka Gyaan...."

* Patterns always get printed rowwise (from left to right)

* Outer loop runs for the no. of rows

  &

  Inner loop runs for the no. of columns

Patterns Halwa Recipe

Steps -

1. First find out the total no. of rows in the pattern.
2. Find the tasks to be performed in each row of pattern.
3. Perform all the tasks of the current row.
4. Do preparation of the next row & change the line.

Home Work - Question

28. $n = 5$

	1	2	3	4		6			
	0	0	0	0	0	0	0	0	0
	0	1	1	1	1	1	1	1	0
	0	1	2	2	2	2	2	1	0
	0	1	2	3	3	3	2	1	0
	0	1	2	3	4	3	2	1	0
	0	1	2	3	3	3	2	1	0
row 7	0	1	2	2	2	2	2	1	0
	0	1	1	1	1	1	1	1	0
	0	0	0	0	0	0	0	0	0

total_rows

① Rows → $2*n-1$

② Digits → $2*n-1$

③ —

—, ✓

x

y

6

2

$(x-y)$

y

x

$t = row-1$

$b = totalrows - row$ $\quad 9-6=3$

$l = col-1$

$r = Digits - col$

$\min(t, b, l, r) \rightarrow$ distan

$\min(5,5,3,3)$

3

amazon

HackerRank

Question:

https://www.hackerrank.com/challenges/printing-pattern-2/problem

29.

$n = 5$

$n = 5$

5	5	5	5	5	5	5	5	5
5	4	4	4	4	4	4	4	5
5	4	3	3	3	3	3	4	5
5	4	3	2	2	2	3	4	5
5	4	3	2	1	2	3	4	5
5	4	3	2	2	2	3	4	5
5	4	3	3	3	3	3	4	5
5	4	4	4	4	4	4	4	5
5	5	5	5	5	5	5	5	5

$5 \to 0$

$4 \to 1$

$3 \to 2$

$2 \to 3$

$n - \min(t, b, l, r)$

$5 - 0 = 5$

$5 - 1 = 4$

$5 - 2 = 3$

$5 - 3 = 2$

LeetCode

118. Pascal's Triangle

https://leetcode.com/problems/pascals-triangle/description/

119. Pascal's Triangle II

https://leetcode.com/problems/pascals-triangle-ii/description/

Pascal's Triangle

30. $n = 5$

(r) 0 1 2 3

(n) 0 1 2 3 4 5

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

1 5 10 10 5 1

$11^0 \rightarrow 1$

$11^1 \rightarrow 11$

$11^2 \rightarrow 121$

$11^3 = 1331$

$^4C_2 = 6$

① Rows → n

② Digits → row

1
2
3

$^nC_r = {}^2C_1 = \frac{2!}{1! \times 1!} = 2$

③ —

④ —

$^5C_3 = \frac{5!}{2! \times 3!} = \frac{\cancel{120}\ 10}{\cancel{2 \times 6}}$

$$^nC_r = \frac{n!}{(n-r)! * r!}$$

(Combination)

$$n! = n \times (n-1) \times \cdots \times 3 \times 2 \times 1$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$= 5 \times 4! = 5 \times 4 \times 3!$$

$$[0! = 1! = 1]$$

Making Halwa -

Kaju Pista Badam Kishmish

1. KP
2. KB
3. KK
4. PB
5. PK
6. BK

$$^4C_2 \Rightarrow \frac{4!}{(4-2)! * 2!}$$

$n \rightarrow 4$, $r \rightarrow 2$

$$6 = \frac{4!}{2! * 2!} = \frac{24}{4} = 6$$

$$^7C_3 = \frac{7!}{4! \times 3!} = \frac{7 \times 6 \times 5 \times \cancel{4!}}{\cancel{4!} \times 3!} = \frac{7 \times 6 \times 5}{3 \times 2 \times 1}$$

$$\frac{\overset{x}{7!}}{\underset{y}{5!}} = \frac{7 \times 6 \times \cancel{5!}}{\cancel{5!}} = 42$$

$$^4C_2 = \frac{4 \times 3}{2 \times 1} \rightarrow \frac{4!}{2! \times 2!} = \frac{4 \times 3 \times \cancel{2!}}{\cancel{2!} \times 2!} = \frac{4 \times 3}{2 \times 1}$$

$$^8C_3 = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} \rightarrow \frac{8!}{5! \times 3!} = \frac{8 \times 7 \times 6 \times \cancel{5!}}{\cancel{5!} \times 3!}$$

$$^{10}C_2 = \frac{10!}{\cancel{8! \times 2!}} \rightarrow \frac{10 \times 9}{2 \times 1}$$

$^8C_3 \rightarrow \frac{8 \times 7 \times 6}{1 \times 2 \times 3}$

$r = 3$

$n = 8$

$\downarrow$

$9 - 2$

$9 - 3$

$i = 3$

for $(i = 1 \rightarrow r)$

$i = 2$

ans $*/ = (n+1-i)$

8×7

ans $/= i$

Important Relation :

	0	1	2		
0	1				
1	1	1			
2	1	2	1		
3	1	3	3	1	
	1	4	6	4	1

$${}^{1}C_0 = \frac{1!}{1! \times 0!} = 1$$

$${}^{n}C_{r+1} = \left(\frac{n-r}{r+1}\right) * {}^{n}C_r$$

$$\left[\begin{array}{l} n \rightarrow \text{row} \\ r \rightarrow \text{col} \end{array}\right]$$

$$\left(\frac{3-2}{2+1}\right) * 3 = 1$$

$$1 \rightarrow 3 \rightarrow \boxed{3 \rightarrow 1}$$

${}^{n}C_r \rightarrow {}^{3}C_1$

${}^{n}C_{r+1} \rightarrow {}^{3}C_2$

Derivation :

$$^nC_r = \frac{n!}{(n-r)! \times r!} \quad - ①$$

$$^nC_{r+1} = \frac{n!}{(n-(r+1))! \times (r+1)!} \quad - ②$$

$$\boxed{^nC_{r+1} = \left(\frac{n-r}{r+1}\right) {}^nC_r}$$

① / ②

$$\frac{^nC_r}{^nC_{r+1}} = \frac{\cancel{n!}}{(n-r)! \times r!} \times \frac{(n-r-1)! \times (r+1)!}{\cancel{n!}}$$

$$= \frac{\cancel{(n-r-1)!} \times (r+1)\cancel{r!}}{(n-r) \times \cancel{(n-r-1)!} \times \cancel{r!}}$$

$$= \frac{(r+1)}{(n-r)}$$